JOHAN DAVID RODRIGUEZ CASTRO

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PROFESSIONAL PROFILE

Final-semester Systems and Software Engineering student (UPB Bucaramanga) working at the intersection of security engineering and applied machine learning. I ship tools rather than findings: an open-source supply-chain scanner, a phishing classifier that audited its own benchmark and published the honest number instead of the flattering one, and a vulnerability prioritisation dashboard that measured away the advantage it was built to demonstrate. Equally comfortable in cloud infrastructure — AWS, Terraform, CI/CD — and full-stack delivery, with Scrum leadership experience and verified C2 English (EF SET 80/100). Seeking a Security, Detection Engineering or Data & AI role.

TECHNICAL SKILLS

Security & Detection: Software supply-chain security (slopsquatting), detection-as-code, MITRE ATT&CK mapping, risk-based vulnerability prioritisation (CISA KEV / EPSS / CVSS), phishing and URL analysis, TOTP 2FA, secure SDLC and CI security gates

Data, ML & AI: LightGBM, XGBoost, scikit-learn; data-leakage detection and honest evaluation design; SHAP explainability; grouped, temporal and walk-forward validation; probability calibration; FinBERT NLP; CrewAI multi-agent orchestration; RAG with ChromaDB; pandas / NumPy

Cloud & DevOps: AWS (IAM, VPC, EC2, Auto Scaling, ALB, RDS Multi-AZ, ECR, CloudWatch), Terraform (IaC), Docker, GitHub Actions CI/CD, k6, Google Cloud (BigQuery ML, Cloud Storage, DLP)

Programming: Python, TypeScript / JavaScript, SQL, Dart, HCL, LaTeX

Databases: PostgreSQL, MySQL, SQLite, DuckDB, ChromaDB; hybrid SQL / NoSQL architectures

Methods & Tools: Agile / Scrum, spec-driven development, pytest, Playwright, mypy strict, UML modelling, Git

EXPERIENCE

Developer / Technical Support — Jan 2026 – Present | Pontificia Universidad Bolivariana

GOATGuard: github.com/Yoyagm/goatguard-app

- Designed and built GOATGuard, a full-stack network security monitoring platform (FastAPI + PostgreSQL + Flutter), to close mobility gaps for administrators managing distributed institutional assets.
- Built an automatic classification engine for network assets (routers, printers, cameras) with real-time monitoring and KPI dashboards, and proactive detection of port scans and intrusion attempts, applying Blue Team and detection engineering principles.
- Hardened the mobile client with a full TOTP 2FA flow (QR enrolment, backup and recovery codes) and JWT held in OS-level secure storage behind a global 401 interceptor.
- Provide technical support to end users across the institution.

Scrum Master — Jun 2025 – Nov 2025 | Pontificia Universidad Bolivariana

- Managed the software development lifecycle under Scrum, facilitating ceremonies and optimising the Product Backlog to ensure incremental value delivery.
- Contributed to the design of a high-availability infrastructure targeting 100,000 concurrent users with a sub-500 ms latency objective, on hybrid SQL / NoSQL / in-memory data architectures.

Assistant, Sports Events — May 2025 – Present | EM Tournaments and Competitions

- Support the organisation and execution of sports tournaments and competitive events.

KEY TECHNICAL PROJECTS

SlopGuard — *Open-Source Supply-Chain Security Scanner* — 2026 | Independent

github.com/Yoyagm/slopguard

- Built a pre-install guard that scores PyPI and npm dependencies against hallucinated, typosquatted and malicious names *before* any code executes, using a five-layer detection engine with zero runtime dependencies.
- Designed a provable anti-false-positive invariant: the opt-in LLM hallucination layer is structurally unable to block a legitimate package on its own.
- Enforced correctness with a CI gate at $\geq$90 % global coverage ($\geq$95 % on critical paths) and eight import-linter architecture contracts, and shipped four integration points from one core: CLI, pre-commit hook, GitHub Action and self-hostable SaaS.

Phishing URL Detector — *Explainable Detection & Dataset Leakage Audit* — 2026 | Independent

yoyagm.github.io/phishing-url-detector | github.com/Yoyagm/phishing-url-detector

- Audited PhiUSIIL (UCI id 967, 235,795 URLs) before training and measured two independent label leaks: a one-line rule scores 99.67 % accuracy, and hand-rolled lexical features on the raw URL — the obvious fix — still score 99.51 % while measuring nothing.
- Retrained on canonicalised host names only and published the honest result: ROC-AUC 0.937 and 70.5 % recall at a 1 % false-positive rate, against phishing submitted two years after the training data was collected.

- Reimplemented exact path-dependent TreeSHAP in dependency-free JavaScript, parity-tested at 1e-12 against the Python reference over 28,917 attributions, so the live demo explains every verdict with no backend.
- Delivered 113 tests — including leakage guards that ship with planted-leak negative controls — a containerised FastAPI service and a nine-page technical report in LaTeX.

Vulnerability Prioritisation Dashboard — *CISA KEV + EPSS + NVD* 2026 | Independent

github.com/Yoyagm/vuln-prioritization-dashboard

- Built a reproducible ingestion pipeline over three public feeds ranking 359,399 live CVEs, with committed data contracts and schema tests that fail the build when a feed changes shape.
- Reproduced the industry claim that EPSS is 6.1× more efficient than CVSS, then showed — with scores frozen at five past cutoffs and only later-confirmed exploitation counted — that most of it is evaluation artefact: EPSS wins the head of the queue (1.2–1.8×) and loses the long tail (0.5–0.7×).
- Bootstrapped 95 % confidence intervals over the positives and guarded the tie-averaging in the core metric with a test that ships with a negative control; 76 tests and five ADRs.

AWS High-Availability Infrastructure — *Terraform, FastAPI & CI/CD* 2026 | UPB

github.com/Yoyagm/eval-final-dgiti

- Codified a three-tier architecture across two availability zones entirely in Terraform: VPC, ALB, Auto Scaling Group, RDS PostgreSQL Multi-AZ, ECR and CloudWatch.
- Enforced least privilege with a strict security-group chain — the database has no public access and no egress, and is reachable only from the application tier.
- Automated deployment with GitHub Actions (lint, test, build, push to ECR, ASG instance refresh), keeping Terraform out of the hot path, and validated load balancing under k6 load against CloudWatch metrics.

Autonomous Market Intelligence System — *Applied ML & Multi-Agent AI* 2025 – Present | Independent

Private repository — walkthrough available on request

- Built an event-driven pipeline integrating market-data APIs, technical-indicator computation and a FinBERT sentiment model, orchestrating four CrewAI agents into daily automated briefings.
- Engineered a governance layer of seven risk checks plus a portfolio drawdown circuit breaker; medium-confidence signals escalate to human approval over Telegram and expire unanswered.
- Trained an XGBoost / RandomForest / LogisticRegression ensemble with walk-forward validation, a purged 21-day gap and Platt calibration; validated by 505 automated tests across 17 suites.
- Runs on Alpaca paper trading and deliberately stays there: a GO/NO-GO gate blocks live capital until the model clears 55 % accuracy (currently 52.9 %).

GreenPulse — *Offline-First Agronomic Monitoring Platform* 2025 | Academic Capstone

- Migrated and redesigned the backend to FastAPI with an offline-first SQLite sync layer and a 35-species plant catalogue including QR-code based scanning; documented in IEEE format.

EDUCATION

B.S. in Systems and Software Engineering Feb 2023 – Present | Pontificia Universidad Bolivariana

- Final-semester student. Advanced coursework in software architecture, system design and industry best practices.
- AWS Academy Cloud Foundations coursework — labs 1–6 completed: IAM, VPC, EC2, EBS, Auto Scaling and Load Balancing, with Terraform IaC implementations (2026).

CERTIFICATIONS

All credentials are publicly verifiable; full list with links at linkedin.com/in/johan-rodriguez-97bb29323.

- **Google Cloud** (2026) — Secure Lakehouse Data • Protect Sensitive Data with Data Loss Prevention • Create ML Models with BigQuery ML • Use APIs to Work with Cloud Storage • Orchestrate Multi-agent Workflows with Gemini Enterprise • Create Your First Gemini Enterprise Application • Build a Smart Cloud Application with Vibe Coding and MCP • Google DeepMind: Train A Small Language Model
- **Cisco Networking Academy** (2026) — Introduction to Cybersecurity
- **Hugging Face** (2026) — AI Agents Course
- **freeCodeCamp** (2026) — Machine Learning with Python • Data Analysis with Python • Scientific Computing with Python • Data Visualization • JavaScript Algorithms and Data Structures
- **EF SET** (2026) — English Certificate, CEFR C2 (Proficient), overall 80/100 (reading 84, listening 89)

LANGUAGES

Spanish — Native | English — C2 Proficient (EF SET 80/100, verified)